

$$x^2$$

Bias - Variance ( $N=2$ ) - Target:  $x^2$

Model Constant Model

$$① \hat{g}^{(D)}(x) = \frac{y_1 + y_2}{2} = x_1^2 + x_2^2$$

$$② \bar{g}(x) = E_{x_1, x_2} \left[ \frac{x_1^2 + x_2^2}{2} \right] = \frac{1}{2} \left( \frac{1}{3} + \frac{1}{3} \right) = \frac{1}{3}$$

● Bias<sup>2</sup>

$$\text{Bias}^2 = \frac{1}{2} \int_{-1}^1 \left( \frac{1}{3} - x^2 \right)^2 dx = \frac{4}{45} \approx 0.0889$$

$$\text{Variance} = E_{x_1, x_2} \left[ \left( \frac{1}{2} (x_1^2 - \frac{1}{3}) + \frac{1}{2} (x_2^2 - \frac{1}{3}) \right)^2 \right] = \frac{2}{45}$$

$$\approx 0.044$$

$$E_{\text{out}} = \text{Bias}^2 + \text{Variance}$$

$$= \frac{4}{45} + \frac{2}{45} = \frac{6}{45}$$

$$E_{\text{out}} \approx 0.1333 \quad \#$$

$$E_{\text{out}}(\text{sim}) \approx 0.1346 \quad \#$$

## Linear Model $g(x)$

Bias<sup>2</sup>:

$$\text{Bias}^2 = \frac{1}{2} \int_{-1}^1 (0 - x^2)^2 dx = \frac{1}{2} \int_{-1}^1 x^4 = \frac{1}{5} = 0.20$$

Variance:

$$\text{Variance} = E_x [E_{x_1, x_2} [(x_1 + x_2)x - x_1 x_2]^2]$$

เพราะ  $x_1$  และ  $x_2$  ไม่สามารถหาความสัมพันธ์กับ  $x$  ได้

$$\text{Variance} = E_x \left[ \frac{2}{3} x^2 + \frac{1}{9} \right] = \frac{2}{3} \left( \frac{1}{3} \right) + \frac{1}{9} = \frac{2}{9} \approx 0.3333$$

$$\begin{aligned} \text{MSE} &= \text{Bias}^2 + \text{Variance} \\ &= 0.2000 + 0.3333 \end{aligned}$$

$$\begin{aligned} \text{MSE} &= 0.5333 \quad \# \\ \text{MSE (sim)} &= 0.5342 \end{aligned}$$

## Linear Through origin

$$\text{Bias}^2 = \frac{1}{2} \int_{-1}^1 (0 - x^2)^2 dx = \frac{1}{5} = 0.2000$$

Variance

$$= E_x [x^2] \cdot E_{x_1, x_2} \left[ \left( \frac{x_1^2 + x_2^2}{x_1 + x_2} \right)^2 \right]$$

$$= \frac{1}{3} \times 0.3441 \approx 0.1147$$

$$\begin{aligned} \text{MSE} &= \text{Bias}^2 + \text{Variance} \\ &= 0.2000 + 0.1147 \\ &= 0.3147 \quad \# \end{aligned}$$

$$\text{MSE (sim)} = 0.3185 \quad \#$$

Sin  $\pi x$

Constant Model

$$\text{Bias}^2 = \frac{1}{2} \int_{-1}^1 (0 - \sin(\pi x))^2 dx = \frac{1}{2} \int_{-1}^1 \sin^2(\pi x) dx = 0.5000$$

$$\begin{aligned} \text{Variance} &= E_{x_1, x_2} \left[ \left( \frac{\sin(\pi x_1) + \sin(\pi x_2)}{2} \right)^2 \right] = \frac{1}{4} \left( \frac{1}{2} \right) + 0 + \frac{1}{4} \left( \frac{1}{2} \right) \\ &= 0.2500 \end{aligned}$$

$$E_{\text{out}} = \text{Bias}^2 + \text{Variance}$$

$$= 0.5000 + 0.2500$$

$$E_{\text{out}} = 0.7500 \quad \#$$

$$E_{\text{out}}(\text{sim}) = 0.7483 \quad \#$$

Linear Model

$$\text{Bias}^2 = \frac{1}{2} \int_{-1}^1 (0.783x - \sin(\pi x))^2 dx \approx 0.2060$$

$$\text{Variance} = E_{x_1, x_2} [(g^{(D)}(x) - \bar{g}(x))^2] \approx 1.6703$$

$$E_{\text{out}} = \text{Bias}^2 + \text{Variance}$$

$$= 0.2060 + 1.6703$$

$$= 1.8763 \quad \#$$

$$E_{\text{out}}(\text{sim}) = 1.8835$$

Linear through origin ( $g(x) = ax$ )

$$\text{Bias}^2 = \frac{1}{2} \int_{-1}^1 (0.79x - \sin(\pi x))^2 dx \approx 0.2718 \quad \#$$

$$\text{Variance} = E_{x_1, x_2} \left[ \left( \frac{x_1 \sin(\pi x_1) + x_2 \sin(\pi x_2)}{x_1^2 + x_2^2} \right) x - 0.79x \right)^2 \right] \approx 0.2372 \quad \#$$

$$E_{\text{out}} = \text{Bias} + \text{Variance}$$

$$= 0.2718 + 0.2372$$

$$= 0.5090 \quad \#$$

$$E_{\text{out}}(\sin) = 0.5135 \quad \#$$